

CATERPILLAR® MACHINE DIAGNOSTICS

INDUSTRIAL INSPECTION & SYSTEM CONFIGURATION AUDIT

Machine ID:	CAT-GEN-C175	Date of Inspection:	2026-07-19
Machine Name:	CAT Diesel Generator	Lead Engineer:	John Administrator
Manufacturer:	Caterpillar Inc.	Model / Category:	C175-16 / CAT Diesel Generator
Diagnostic Status:	Fault (Score: 0%)	Engine Version:	v2.4-deterministic

System Hardware & Firmware Configuration Audit

Below is a side-by-side comparison of the active telemetry and local software modules against original factory reference parameters.

Parameter	Factory Reference	Current Machine	Audit Status
Firmware Version	v9.4.0	v9.0.2	MISMATCH (CRITICAL)
PLC Version	v8.32	v8.00	MISMATCH (WARNING)
CPU Architecture	Intel Core i3-7100U	Intel Pentium 4415U	MISMATCH (CRITICAL)
RAM (Memory)	16GB DDR4	16GB DDR4	MATCHED
Storage	128GB SSD	128GB SSD	MATCHED
Sensor Count	24	24	MATCHED
Communication Ports	USB, COM1, COM2, Ethernet, Modbus-TCP, Profibus	COM1, Ethernet, Modbus-TCP, USB	MISMATCH (CRITICAL)
Installed Modules	Analog Input, Digital IO, CAN Bus controller, Power Logger Card	Analog Input, CAN Bus controller, Digital IO	MISMATCH (CRITICAL)
Operating Temperature	Under 70 C	44.5 C	MATCHED
Power Supply Status	Stable	Stable	MATCHED

Detected Mismatches & Anomalies

1. Firmware Version Mismatch [CRITICAL]

Firmware version mismatch. Expected 'v9.4.0', found 'v9.0.2'.

2. PLC Version Mismatch [WARNING]

PLC configuration version mismatch. Expected 'v8.32', found 'v8.00'.

3. CPU Architecture Mismatch [CRITICAL]

Processor type mismatch. Reference specifies 'Intel Core i3-7100U' but found 'Intel Pentium 4415U'.

4. Communication Ports Mismatch [CRITICAL]

Required port(s) offline: Profibus, COM2.

5. Installed Modules Mismatch [CRITICAL]

Critical expansion module(s) missing or failed: Power Logger Card.

AI-Powered Diagnostics Analysis

Machine Safety & Health Assessment

DEGRADED / ACTION REQUIRED (Health: 0%): Mismatches or anomalous sensor readings detected. Operational risks present.

Root Cause Analysis

The diagnostic engine flagged the following configuration mismatches: - Firmware Version: Firmware version mismatch. Expected 'v9.4.0', found 'v9.0.2'. - PLC Version: PLC configuration version mismatch. Expected 'v8.32', found 'v8.00'. - CPU Architecture: Processor type mismatch. Reference specifies 'Intel Core i3-7100U' but found 'Intel Pentium 4415U'. - Communication Ports: Required port(s) offline: Profibus, COM2. - Installed Modules: Critical expansion module(s) missing or failed: Power Logger Card. Typical root causes include: field component replacement with unauthorized modules, out-of-date PLC programming flashes, firmware regression during maintenance cycles, or thermal cooling degradation.

Suggested Maintenance Worksteps

1. Connect Service Tool (CAT ET or Siemens TIA Portal) and flash target firmware and PLC image profiles matching reference configurations.
2. Audit active hardware modules. Replace generic CPU boards, RAM modules, or expansion cards with OEM-authorized parts.

Crucial Safety Notes

- Verify power source disconnect before opening PLC enclosures.
- Ensure constant power backup is connected to the PLC logic controllers during the programming and flashing sequence.
- Follow Electrostatic Discharge (ESD) wrist strap procedures when handling exposed logic boards or memory chips.

Verification & Troubleshooting Steps

1. Run firmware CRC checksum validation command. Check serial console output for boot loader warnings.
2. Perform full system diagnostic self-test from the service shell. Verify device tree registration of all local modules.

PREPARED BY:

Automated Diagnostic Engine
CAT Diagnostic Systems

REVIEWED & SIGNED BY:

Lead Engineer: John Administrator
CAT Certified Field Inspector